

The following problems will help you to prepare for the final exam. They indicate the style (but not the amount!) of problems that will form the bulk of the final: 1. giving examples, 2. stating/correcting important definitions, 3. stating/correcting and illustrating important results, 4. proving smaller results, 5. ASN, 6-10. free response questions, mostly similar in content, style, and difficulty to the HWs and Homeworks. You should also be prepared for questions like those on the Midterm Review Problems and from the midterm itself; you should also be prepared to use all of the skills from the skills quizzes. A key for these review problems will be posted in Files.

1. Give an example of each of the following, or explain why it is not possible.

- a) A nilpotent matrix of rank 2.
- b) An orthogonal matrix that is nilpotent.
- c) A pair of vectors in $\mathbb{R}^3$ such that their span is a plane.
- d) A pair of orthogonal vectors in $\mathbb{R}^3$ such that their span is a plane.
- e) A pair of vectors in $\mathbb{R}^3$ such that their span is a line.
- f) A pair of orthogonal vectors in $\mathbb{R}^3$ such that their span a line.
- g) An orthonormal basis for the nullspace of $\begin{pmatrix} 4 & 6 & 10 \end{pmatrix}$.
- h) A non-zero matrix satisfying the polynomial equation $x^4 + x = 0$.
- i) An invertible matrix satisfying the polynomial equation $x^4 + x = 0$.
- j) A non-zero matrix satisfying the polynomial equation $x^4 = 0$.
- k) An orthogonal matrix satisfying the polynomial equation $x^4 = 0$.
- l) An invertible matrix satisfying the polynomial equation $x^4 = 0$.
- m) Two non-equal matrices, both with characteristic polynomial $\lambda^2 - 5\lambda + 6$.
- n) A matrix that is diagonal but not diagonalizable.
- o) A matrix with 4 distinct eigenvalues.
- p) An orthogonal matrix that is not invertible.
- q) An inner product on $\mathbb{R}^2$ that is not the standard inner product.
- r) Two vectors such that $\|v\| = \|w\| = 1$ and $v \cdot w = 1/2$.
- s) Two vectors such that $\|v\| = \|w\| = 1$ and $v \cdot w = 2$.
- t) A square matrix with eigenvalues 1, 2, and 3.
- u) A non-square matrix with eigenvalues 1, 2, and 3.
- v) A square matrix with singular values 1, 2, and 3.
- w) A non-square matrix with singular values 1, 2, and 3.
- x) A non-diagonal matrix similar to a diagonal matrix.
- y) A non-identity matrix similar to an identity matrix.

- z) A non-orthogonal matrix similar to an orthogonal matrix.
- aa) An orthogonal matrix with determinant -1.
- bb) An orthogonal matrix with determinant 10.
- cc) An upper triangular matrix that is not a Jordan block.
- dd) An invertible matrix that is a Jordan block.
- ee) A unitary matrix and a non-unitary matrix that are row equivalent.
- ff) A matrix with eigenvalues 4 and $2 + i$.
- gg) A real matrix with spectrum that contains 4 and $2 + i$.
- gg') A real matrix with spectrum that equals 4 and $2 + i$.
- hh) A matrix with spectrum that contains 5 and 6 and that has determinant 20.
- ii) A matrix that has only 5 and 6 as eigenvalues and that has determinant 20.
- jj) A matrix that has only 2 and 5 as eigenvalues and that has determinant 20.
- kk) A nilpotent matrix with determinant 9.
- ll) An invertible matrix with determinant 9.
- mm) An orthogonal matrix with determinant 9.
- nn) An orthogonal matrix with eigenvalues 3 and $1/3$.
- oo) A 2×2 matrix that is not in Jordan canonical form.
- pp) Two matrices, each with nullity 2, whose product has nullity 3.
- qq) Two matrices, each with nullity 2, whose product has nullity 4.
- rr) Two matrices, each with nullity 2, whose product has nullity 5.
- ss) A matrix with 2 real and 2 non-real eigenvalues.
- tt) A real matrix with no real eigenvalues.
- uu) A complex matrix with no complex eigenvalues.
- vv) A real number that is never the eigenvalue of a real matrix.
- ww) A real matrix that is not similar to a real upper triangular matrix.
- xx) A complex matrix that is not similar to a complex upper triangular matrix.
- yy) A matrix with trace 0 and determinant 1.
- zz) A matrix with trace 0 and eigenvalues 2 and 3.
- A) A square matrix A for which there exists no square matrix B where $B^2 = A$.
- B) Two matrices A and B that commute but whose squares A^2 and B^2 do not commute.
- C) Two matrices A and B that do not commute but whose squares A^2 and B^2 commute.
- D) An orthonormal basis for $\mathbb{R}^2$ that contains no standard basis vectors.
- E) An orthonormal basis for $\mathbb{R}^3$ that contains exactly two standard basis vectors.
- F) A nilpotent permutation matrix.

- G) An invertible permutation matrix.
- H) A permutation matrix with trace 0.
- I) A basis for the subspace of $M_{2 \times 2}^{\mathbb{R}}$ consisting of symmetric matrices.
- J) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 2.
- K) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 4.
- L) A matrix with an eigenvalue λ with algebraic multiplicity 4 and geometric multiplicity 0.
- M) Two matrices with different ranks but the same nullspace.
- N) Two matrices with different ranks but the same nullity.
- O) Two square matrices with different ranks but the same nullity.
- P) A square matrix A and its $D + N$ decomposition.
- Q) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector.
- R) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector with eigenvalue -1.
- S) A non-diagonal matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector with eigenvalue -1.
- T) A matrix that has $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ as an eigenvector but where $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ is not an eigenvector.
- U) A matrix whose minimal polynomial is equal to its characteristic polynomial.
- V) A 3×3 matrix whose minimal polynomial is $(1 - \lambda)(4 - \lambda)$.
- W) A matrix with characteristic polynomial $\lambda^2 - 7\lambda + 12$ and minimal polynomial $\lambda - 1$.

2. Explain what is wrong about the following definitions and correct them.

A vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$.

In an inner product space, a set of vectors is **orthonormal** if each pair of distinct vectors is orthogonal (so that the inner product of the pair is 0).

3. Complete the following:

- a) State the Diagonalization Theorem. Illustrate the theorem with an example.
- b) Give four examples of conditions that are equivalent to a square matrix being invertible.
- c) State the Cayley–Hamilton Theorem. Illustrate the theorem with an example.
- d) Correct the following theorem statement:

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a normal matrix and T is an upper triangular matrix.

4. Prove each of the following:

- a) If λ is an eigenvalue of a square matrix A , then $\det(\lambda I - A) = 0$.
- b) Use the properties of an inner product to show that the following function $\$$ on the vector $\mathbb{R}^2$ is not an inner product: $\begin{pmatrix} a \\ b \end{pmatrix} \$ \begin{pmatrix} c \\ d \end{pmatrix} = ac - bd$.
- c) If A is a real 2×2 matrix with eigenvalues 1 and -1 , then $A^2 = I$.
- d) For any non-constant real polynomial p , there exists a complex matrix A where $p(A) = 0$.

5. Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

- a) A symmetric matrix has all real eigenvalues.
- b) For a nilpotent matrix N , $N^{100} = 0$ where 0 is the zero matrix.
- c) An upper triangular matrix is diagonalizable.
- d) A square matrix A and its transpose A^T have the same eigenvectors.
- e) Two distinct eigenspaces of an operator have a non-zero vector in common.
- f) Two distinct subspaces of a vector space have a non-zero vector in common.
- g) The Euclidean norm of a vector v in $\mathbb{C}^n$ equals the Euclidean norm of the conjugate of v .
- h) A self-adjoint matrix is symmetric.
- i) A self-adjoint matrix is a normal matrix.
- j) A square matrix is equal to a product of elementary matrices.
- k) The sum of two diagonalizable matrices is diagonalizable.
- l) The sum of two orthogonal matrices is orthogonal.
- m) If a matrix A is real, then $A - iI$ is invertible.
- n) If $u \cdot v = 0$ and $v \cdot w = 0$, then $w \cdot u = 0$.
- o) A given square matrix is either invertible or nilpotent.
- p) Two matrices with the same characteristic polynomial are equal.
- q) Two matrices with the same characteristic polynomial have the same determinant.
- r) Orthogonal projection to a plane in $\mathbb{R}^3$ is represented by an orthogonal matrix.
- s) A diagonal matrix is nilpotent.

6. a) Compute the eigenvalues of the following matrix in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$, as well as a basis for each eigenspace:

$$\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$$

b) Compute the eigenvalues (with multiplicity) of the following linear operator on $\mathbb{R}^3$: reflection across the plane $x + 2y + z = 0$.

c) Let D be the linear operator acting on the real vector space of real differentiable functions in a single variable x . For each of the following values λ , give an example of a vector v such that v is an eigenvector under D having eigenvalue λ : 1, 2, 0, 37.

d) Continuing from part c, give an example of (i) a vector that is not an eigenvector for D and (ii) a vector that is not an eigenvector for D but that is an eigenvector for D^2 .

7. a) Find the singular values of each of the following matrices:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 10 \end{pmatrix}$$

8. Compute the 10th power of each of the following matrices. You do not need to fully simplify the entries of the resulting matrix; it is fine if an entry is something like $5 \cdot 4^{10}$. (Hint: use either the Diagonalization Theorem or the $D + N$ decomposition; see the handout on powers of 2×2 matrices.) Also, compute the determinants and eigenvalues of the resulting matrices.

$$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \begin{pmatrix} -3 & 4 \\ -9 & 9 \end{pmatrix}$$

9. Compute a QR decomposition of the following matrix:

$$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix}$$

10. a) Find all vectors in $\mathbb{R}^4$ that are orthogonal to both of the following vectors:

$$\begin{pmatrix} 2 \\ 0 \\ 4 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

b) Find a basis for the subspace in $M_{2 \times 2}^{\mathbb{R}}$ that is orthogonal to the subspace of the symmetric matrices (using the standard inner product). (Looking back at problem 1(I) might help.)